# Dynamic QR Attendance System

**Attendance that expires before it can be reused.**

A lightweight, security-aware attendance workflow for modern classrooms.

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github.com/Abdelrahman-Noaman/dynamic-qr-attendance



# Static attendance creates a trust gap.

Static QR codes and manual attendance waste time and compromise data integrity.

SCREENSHOT + SHARE

**Static codes travel beyond the room.**

MANUAL FRICTION

**Roll calls interrupt classroom flow.**

DUPLICATE RECORDS

**Repeated submissions weaken the dataset.**

NO FEEDBACK

**Students leave unsure if attendance landed.**

# Scan. Validate. Record. Confirm.

A compact workflow with security checks at every handoff.

01

### Rotating token

Refreshes about every 10 seconds

02

### Server validation

Rejects replay and malformed requests

03

### Signed ticket

Short-lived proof for submission

04

### Duplicate control

One reliable record per student

05

### Sheets logging

Familiar, inspectable records

06

### Email confirmation

Immediate student feedback

# One scan. Six controlled steps.

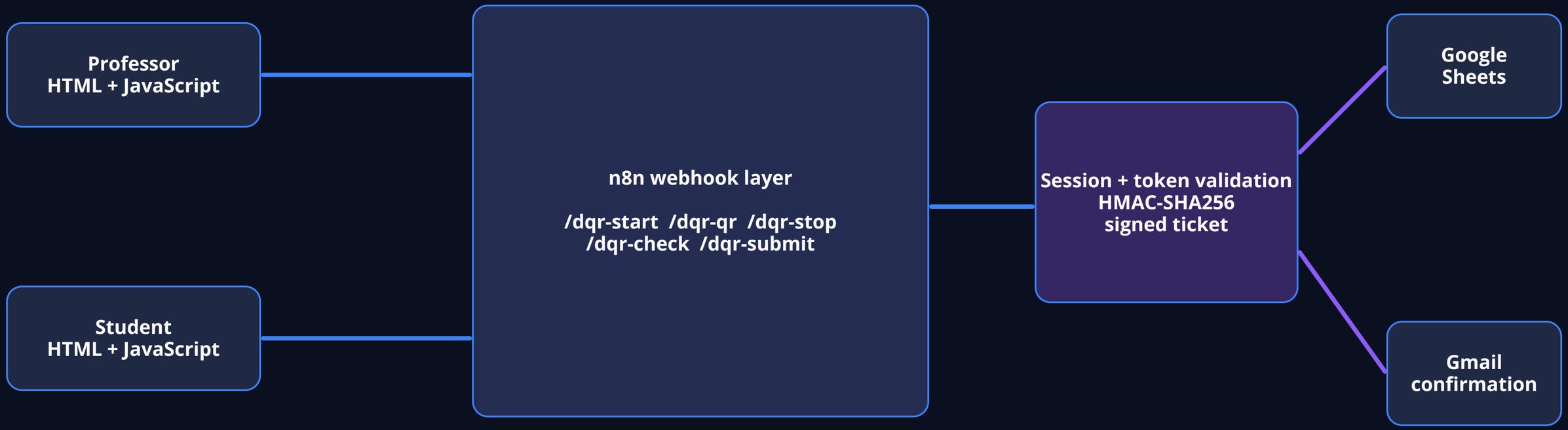



QR rotation ≈ 10 sec

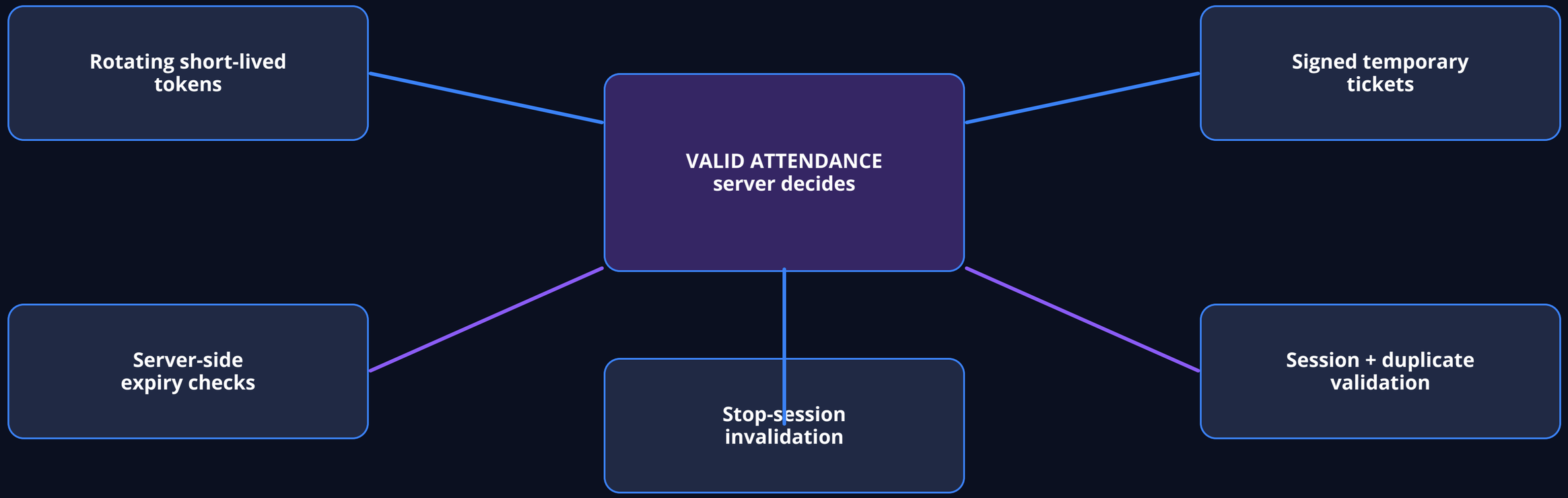
Ticket validity ≈ 3 min

# Technical architecture

Browser clients → n8n orchestration → Google Workspace



# Security by design—not by assumption.




These controls reduce common replay and duplicate-attendance risks—but cannot guarantee physical presence or prevent live QR relay attacks.

# Competitor landscape

A cautious, source-led feature comparison—no pricing or superiority claims.

| Feature | Static QR + Google Forms | QRoll | Lincard | Dynamic QR prototype |
|---|---|---|---|---|
| Token rotation | No | Not verified | Not verified | Yes |
| Backend validation | No | Verified | Not verified | Yes |
| Duplicate controls | Limited | Verified | Not verified | Yes |
| Email confirmation | Optional | Not verified | Not verified | Yes |
| Google Workspace | Yes | Not verified | Not verified | Yes |
| Deployment complexity | Low | Not verified | Not verified | Moderate |

# Small stack. Serious controls.

What makes the prototype distinct is the combination—not a single feature.

01 **Rotating QR + signed ticket**

02 **Session, expiry + duplicate checks**

03 **Sheets records + email confirmation**

04 **Transparent n8n workflow**

05 **Google Workspace—not a campus suite**

# Low friction for both sides of the room.

Lectures • Laboratories • Workshops • Tutorials • Training

STUDENT EXPERIENCE

1 Scan QR

2 Submit attendance

3 Receive confirmation email

PROFESSOR EXPERIENCE

1 Start session

2 Display rotating QR

3 Stop session

4 Review records

# Test what matters. Build what’s missing.

## ENGINEERING TEST MATRIX

| | |
|---|---|
| Valid token submission | PASSED |
| Expired token rejection | TO BE TESTED |
| Invalid signature rejection | TO BE TESTED |
| Duplicate prevention | TO BE TESTED |
| Stopped session rejection | TO BE TESTED |
| Google Sheets logging | PASSED |
| Gmail delivery | PASSED |

## LIMITATIONS

- Sheets-only records
- n8n state; no database yet
- Shared professor key
- No strong presence proof
- No institutional RBAC

## ROADMAP

Database state • Institutional auth
Rate limits + audit logs
Anti-relay controls • Privacy + load testing

# Simple for students.
# Practical for professors.
# More careful than static QR.

Dynamic tokens. Transparent n8n workflows. Google Workspace records. Built for classrooms, labs, workshops, and training.

**EXPLORE • CONTRIBUTE • TEST**

**github.com/Abdelrahman-Noaman/dynamic-qr-attendance**

